

standard_assessment_item_writing

Assessment Item Writing Guidelines

Purpose

Establishes the quality bar for any assessment item — a quiz question, practical exercise, or coding challenge — used to evaluate Junior Backend Developer competency. These guidelines apply equally to items written by a human curriculum designer and items generated by an automated system, since the quality bar for what makes an item usable does not depend on how it was produced.

General Principles

Every assessment item must map to exactly one named competency from an approved competency framework, such as the organization's Backend Competency Framework. Items that do not map cleanly to a specific, named competency should not be used in a formal evaluation, regardless of how well-written the item is in isolation — an unmapped item cannot be meaningfully interpreted as evidence of anything specific about the person being assessed.

Every item must also specify its intended difficulty or proficiency level, aligned to the levels defined in the relevant competency rubric, so that a correct or incorrect response can be interpreted against a specific expected level rather than in the abstract.

Requirements for Multiple-Choice Items

- Exactly one option must be unambiguously correct given the item's stated premise and any constraints provided.
- Distractor options (the incorrect choices) must be plausible but clearly incorrect upon careful reading by someone who genuinely possesses the target competency — distractors should not rely on trick wording, ambiguous phrasing, or obscure edge cases unrelated to the competency being tested.
- Items must not rely on outdated, deprecated, or organization-specific practices as the stated "correct" answer unless the item explicitly frames itself as testing knowledge of the organization's specific conventions rather than general industry practice.

Requirements for Practical and Coding Items

- The expected solution must be verifiable by an automated test wherever feasible, rather than relying solely on manual grading, both for consistency across graders and for scalability as assessment volume grows.
- Instructions must explicitly specify any constraints relevant to correctness or grading — including time or space complexity requirements, disallowed libraries or language features, and expected input/output format — since ambiguity in these areas leads to inconsistent grading even when the underlying competency assessment intent was sound.
- A reference solution and a complete answer key must be produced alongside the item itself, at the time the item is written, not added retroactively after the item has already been used in an evaluation.

Validation Requirement

Before an assessment item is used in a real evaluation, it must pass an automated validation pass checking, at minimum, the following:

1. The stated correct answer is actually correct given the item's premise, and is verifiably so — not merely plausible.
2. The item does not contain factual errors when compared against the source competency framework or reference material it is meant to test.
3. The item's actual difficulty is appropriate for its stated competency level, based on comparison against other validated items at the same stated level.

Items that fail any part of this validation must be revised and re-validated, or discarded entirely — they must never be used as-is with an informal note such as “fix the answer key later,” since a known-flawed item used in a real evaluation produces evaluation results that cannot be trusted, and this risk compounds if the flawed item is reused across multiple evaluation cycles before being caught.

Common Failure Pattern to Avoid

A frequent and specifically dangerous failure mode is a “plausible-but-wrong” item: one that reads as well-formed, reasonable, and professionally written, but contains an incorrect answer key, a flawed premise, or does not actually test the competency it claims to test. This pattern is dangerous specifically because such items are difficult to catch through casual reading — they look correct on the surface — which is exactly why a structured, automated validation pass is required rather than relying on a human reviewer's read-through alone to catch every instance.

Worked Example of a Plausible-But-Wrong Item

Consider a multiple-choice item intended to test understanding of HTTP status codes: “Which status code should be returned when a resource is successfully deleted?” with options including 200, 202, 204, and 404. If the answer key marks 200 as correct, this is plausible (200 does indicate success in the loosest sense) but arguably wrong against the specific competency being tested, since 204 (No Content) is the more precise, conventionally correct answer for a successful deletion with no response body — an evaluator relying on this item without validation could incorrectly mark a well-informed learner's alternative answer as wrong, or reinforce an imprecise understanding in learners who select the flawed “correct” answer.

Owner

Curriculum Design Standards Committee, responsible for reviewing this standard whenever a systemic pattern of flawed items is identified during validation or after use in a real evaluation.